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Configuring Barcode Preprocessing Rules

Overview

In warehouse operations, a wide range of SKUs and suppliers results in barcodes of various formats, structures, and qualifiers. Not all scanned barcodes arrive in a format that matches your Warehouse Management System (WMS) requirements. These variations can lead to inconsistencies during scanning and downstream processing.

The **Barcode Preprocessing (BarcodePreProcView)** UI allows you to configure rules for preprocessing scanned barcodes. You can define barcode matching conditions and specify logic used to transform barcode values. When a scanned barcode meets the defined criteria, the system applies the configured preprocessing rules and updates the barcode before further processing.

Note:

- Rules can be configured only in Redwood UI format.
- Rules are user group-specific.

Processing Flow

When a barcode is scanned, the system processes it in the following order:

1. Match the scanned barcode against configured preprocessing rules (by sequence).
2. Apply the corresponding update logic (if matched).
3. Stop further rule evaluation.
4. Pass the updated barcode for downstream processing.

Key Capabilities

Use barcode preprocessing rules to:

- Standardize barcode formats across suppliers.
- Extract or restructure barcode segments from the scanned barcode.
- Validate barcode input in real time.
- Reject invalid barcodes before they disrupt operations, reducing manual intervention and errors.
- Adapt to new barcode formats without code changes.

Configure Barcode Preprocessing Rules

In the **Barcode Preprocessing** UI, you can set criteria for when to preprocess the scanned barcodes.

To configure barcode preprocessing rules, do the following:

1. Navigate to **Barcode Preprocessing** UI.
2. Click **Create**.
3. On the **Create** form, enter or select the following fields or toggles as required:
    - **Sequence Nbr:** Defines the order in which barcode preprocessing rules are evaluated.
    - **Prefix Text:** Specifies the starting text to identify matching barcodes.
    - **Match Function:** Determines how the system evaluates the barcode. You can select one of the following match functions:

Match Function	Behavior
Min Length	Matches if the scanned barcode meets the minimum length. On selecting this function, enter the minimum length value to match with the scanned barcode length.
Exact Length	Matches if the scanned barcode is of exact specified length. On selecting this function, enter the exact length value to match with the scanned barcode length.
Begins With	Matches if the scanned barcode starts with the configured text. On selecting this function, enter the string to match with the scanned barcode's beginning string.
Contains	Matches if the scanned barcode contains a specific string. On selecting this function, enter the string or text to match with the scanned barcode if it contains the configured text.
Regex Match	Matches if the scanned barcode fits the regular expression provided. On selecting this function, enter a regular expression in the regex field.

Note: All text entries are case-sensitive.

- **Reject Barcode** toggle: When enabled, the system rejects the matching scanned and error is displayed during scanning. You can enable this toggle to reject invalid barcodes.

Note: Either **Sequence Nbr** or **Match Function** is mandatory for each barcode preprocessing rule. You can also configure both fields.

4. Click **Save**.

You can create multiple rules with different sequence numbers.

The system matches the scanned barcodes against barcode preprocessing rules in sequence. Once the scanned barcode matches a rule, any associated update logic is executed and no further rules are evaluated.

Barcode Preprocessing Update rules for a matching sequence define how the barcode is to be updated or handled next.

Configure Barcode Preprocessing Updates

In the **Barcode Preprocessing Updates** screen, you can configure logic to update the scanned barcodes.

To define how a matched barcode should be updated, do the following:

1. In the **Barcode Preprocessing** UI, select a barcode preprocessing rule.
2. Click **More Actions** > **Barcode Preprocessing Updates**.
3. Click **Create**.
4. On the **Create** form, enter or select the following fields:
   - **Sequence Nbr:** Determines update execution order.
   - **Update Function:** Logic to transform the scanned barcode. You can select one of the following update functions:

Update Function	Behavior	Example

Substring	Extracts a portion of the barcode. On selecting this function, enter **startindex** and **endindex** values. It includes the character at the start index and excludes the character at the end index. Index value starts with 0. If the provided end index is greater than the last index of the scanned barcode, the system uses characters up to the position of the last character.	**Scanned Barcode**: ARTSKU1234567 **Startindex:** 3 **Endindex:** 9 **Result:** SKU123 (from 3rd to 8th character)
Strip String	Removes all occurrences of a string. On selecting this function, enter **string value** to remove from the scanned barcode.	**Scanned Barcode**: ARTSSKUARTS123 **Strip String**: ARTS **Result:** SKU123
Strip Regex	Removes parts of the scanned barcode using regex. On selecting this function, enter a regular expression logic in the **regex** field to remove portions of the scanned barcode. You can use the standard regular expression patterns.	**Scanned Barcode**: ARTSKU123456 **regex**: ^(.{3}) **Result:** SKU123456 (first three characters are removed)
Insert At	Inserts a string at the specified position. On selecting this function, enter **Position** and **String to insert** values. The position value starts with 0. If the provided position is more than the last index of the scanned barcode, the system considers last character's position as the position to insert string.	**Scanned Barcode**: ART12345 **Position**: 3 **String to insert**: SKU **Result:** ARTSKU12345 (inserts string at 3rd position).
Regex Replace	Performs advanced transformations using regex groups. On selecting this function, enter a regular expression logic in the **regex** field.	You can use this function to create multi field barcode from the scanned barcode. **Scanned Barcode**: SSBPLPN2201 **regex**: ^SSBP(.{7}) **String to replace**: SSMFB[A]$1! **Result:** SSMFB[A]LPN2201!

5. Click **Save**.

You can create multiple updates with different sequence numbers.

Execution Behavior

Barcode Preprocessing Updates execute in sequence order. Output of one update becomes input for the next.

Examples

Example 1: Extract Substring

Extract a portion from the scanned barcode.

Input Barcode: ARTSSKU1234567

Barcode Preprocessing Rules:

Group	Sequence Nbr	Prefix Text	Reject Barcode	Match Function	Parameters
Group1	1	ART	false	min_length	{“length”:6}

The system matches the scanned barcode ARTSSKU1234567 based on the prefix and match function.

Barcode Preprocessing Update (for Sequence 1):

Barcode Preprocessing Match Rule	Sequence Nbr	Update Function	Configuration
Company/Facility/1	1	substring	startindex = 4 endindex = 10

The update rule extracts characters from position 4 to 9.

After updates: SKU123

Example 2: Reject Invalid Barcode

Reject all barcodes starting with “SSITEM”.

Barcode Preprocessing Rules:

Group	Sequence Nbr	Prefix Text	Reject Barcode
Group2	1	SSITEM	true

If the scanned barcode is SSITEM012XYZ, the system matches the prefix and rejects the barcode.

Example 3: Multi-Step Processing

When a barcode (e.g., **SKUSLPN12345**) is scanned during a mobile transaction, the system evaluates it against each rule in sequence number order until a match is found.

Barcode Preprocessing Rules:

Group	Sequence Nbr	Prefix Text	Reject Barcode	Match Function	Parameters
Group3	1	SKU	false	min_length	{“length”:14}
Group3	2	SKU	false	exact_length	{“length”:20}
Group3	3	SKU	false	begins_with	{“string”:“SKUNR”}
Group3	4	SKU	false	contains	{“contains”:“SKU-SKU”}
Group3	5	SKU	false	regex_match	{“regex”:“S.S”}

For the barcode SKUSNLPN12345, the system evaluates the barcode preprocessing rules in order from sequence 1 to 5:

1. Sequences 1 to 4 do not match with this barcode according to their configuration.
2. Sequence 5 matches this barcode, because the regular expression rule applies (regex_match: {"regex":"S.S"}).

As soon as the sequence 5 matches, the system triggers the associated barcode preprocessing update rules.

Barcode Preprocessing Update (for Sequence 5):

Barcode Preprocessing Match Rule	Sequence Nbr	Update Function	Configuration
Company/Facility/5	1	strip_string	string = SN
Company/Facility/5	2	substring	startindex = 0 endindex = 8

For the barcode SKUSNLPN12345, the system updates the barcode in a sequence from 1 to 2:

1. Sequence 1: Removes all occurrences of "SN". (SKULPN12345)
2. Sequence 2: Extract the first 8 characters from the result. (SKULPN12)

Input barcode: SKUSNPLN12345

After updates: SKULPN12

Example 4: Conditional Extraction Based on Supplier Prefix

You receive SKUs with multiple supplier barcodes, each with a unique prefix and different part number formats:

- Supplier1: Barcodes start with the prefix "S1-", with the part number immediately following (e.g., S1-123456).
- Supplier2: Barcodes start with the prefix "S2", with the part number immediately following (e.g., S2123456).

You can make the system to recognize the supplier based on the prefix and extract the part numbers.

Barcode Preprocessing Rules:

Group	Sequence Nbr	Prefix Text	Reject Barcode	Match Function	Parameters
Group4	1	S1-	false	min_length	{"length":6}
Group4	2	S2	false	min_length	{"length":6}

Barcode Preprocessing Update (for Sequence 1):

Barcode Preprocessing Match Rule	Sequence Nbr	Update Function	Configuration
Company/Facility/1	1	strip_string	String = S1-

Barcode Preprocessing Update (for Sequence 2):

Barcode Preprocessing Match Rule	Sequence Nbr	Update Function	Configuration
Company/Facility/2	1	strip_string	String = S2

For Supplier1 barcode "S1-123456":

- The rule matches the "S1-" prefix.
- The barcode preprocessing update removes the string "S1-". So, the updated barcode is "123456".

For Supplier2 barcode "S2123456":

- The rule matches the "S2" prefix.
- The barcode preprocessing update removes the string "S2". So, the updated barcode is "123456".